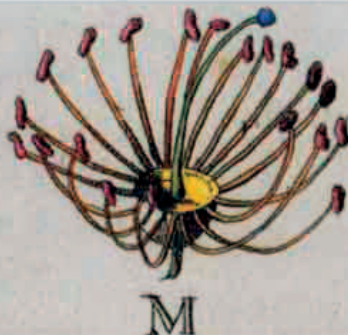
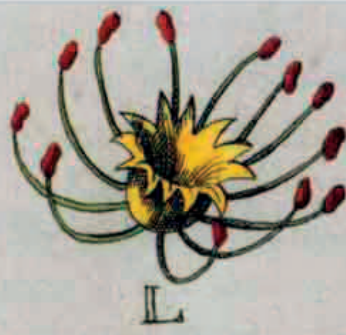
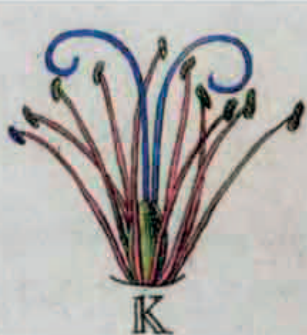




Carl Linnaeus's classification of plants focused on their sexual organs, the pistils and stamens, as illustrated here by botanical artist Georg Ehret, who worked with Linnaeus in the 1730s.

AMERICAN INVENTOR AND POLYMATH BENJAMIN FRANKLIN

(1705–90) had proved in 1751 that lightning is natural electricity. Some two years later, he demonstrated with his new invention, the **lightning rod**, how buildings could be protected against this hazard. The simple device, still used today, comprised a metal rod placed on top of a building to draw down lightning and conduct it harmlessly to the ground through a metal wire, saving the building from damage. Although many argued that drawing lightning only increased the likelihood of a strike, the idea caught on quickly. But Czech inventor **Prokop Diviš** (1698–1765) independently invented a similar device around the same time, and it was Diviš's design that became more widely used.

British naval surgeon **James Lind** published in 1753 the first edition of *A Treatise of the Scurvy*, the result of six years research into the benefits of citrus fruit as a preventive of the dread shipboard disease. It was several decades before anyone acted fully on Lind's theory.

Lightning conductor

Franklin's lightning rod initially met with worried opposition, but soon many buildings were sprouting these "property protectors."

Swiss mathematician **Leonhard Euler** (1707–83) addressed the question of how three heavenly bodies, such as the Sun, Moon, and Earth, interact. He approached what is known as the three-body problem in his book *Theoria Motus Lunae (A Theory of Lunar Motion)*, and eventually found a solution in 1760. Euler also pioneered studies into how the gravitational pull between the Moon and Earth drives tides on Earth. Understanding of such forces was still in its infancy in 1754, when Dutch dike supervisor **Albert Brahms** (1692–1758) began the first scientific recordings of tide levels.

Ruder Josip Bošković (1711–87) of Dubrovnik, who made significant contributions to at least half a dozen scientific fields, claimed that the Moon has no atmosphere. In fact, we now know that the Moon does have a sparse atmosphere, although not enough to support

life as known on Earth. But Bošković's almost-correct theory was a key step in the process of understanding other worlds beside our own.

Scottish chemist **Joseph Black** (1728–99) also found out something about Earth's atmosphere with his discovery of **carbon dioxide**, which he called "fixed air." Black learned that this gas is heavier than air, does not allow flames to burn, can cause asphyxiation, and is exhaled by animals in breath.

During this period, Swedish naturalist **Carl Linnaeus** (see 1737–39) produced his work on classifying plants, *Species Plantarum (The Species of Plants)*, which covered some 6,000 plants and gave each one a binomial (two-part) Latin name indicating genus and species. This system provides the basis for plant nomenclature used by botanists today.

30,000
ESTIMATED
NUMBER OF
AMPS CARRIED
IN A BOLT OF
LIGHTNING

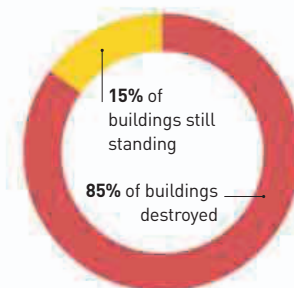
“...SYSTEMS OF MANY STARS, WHOSE DISTANCE PRESENTS THEM IN SUCH A NARROW SPACE THAT THE LIGHT... REACHES US, IN A UNIFORM PALE GLIMMER...”

Immanuel Kant, German philosopher in *Universal Natural History and Theory of the Heavens*, 1755

AROUND MIDMORNING ON NOVEMBER 1, 1755

the city of Lisbon in Portugal was devastated by an earthquake now estimated at magnitude 8.5 on the Richter scale (see 1935). After studying the after-effects, British geologist **John Michell** (1724–93) correctly suggested that earthquakes travel as **seismic waves**, which alternately compress and stretch the ground. Michell worked out that the quake's epicenter—the point where the waves that shook Lisbon originated—was in the eastern Atlantic between the Azores and Gibraltar.

British engineer **John Smeaton** (1724–92) improved the stability of buildings with his pioneering



Great quake

Many major buildings and around 12,000 dwellings were destroyed by Lisbon's earthquake of 1755.

use of hydraulic lime, a cement that sets underwater and is resilient to deterioration when wet. Smeaton used hydraulic lime to build the third Eddystone lighthouse off the coast of southwest England.



1753 In U.S., Benjamin Franklin invents lightning rod

May 1, 1753 Carl Linnaeus publishes his system of plant classification

1753 British naval officer James Lind publishes *A Treatise of the Scurvy*

1754 Swiss polymath Charles Bonnet pioneers the idea that the physical mind is key to personality

1755 German philosopher Immanuel Kant develops nebular theory for formation of the Solar System

1753 Swiss mathematician Leonhard Euler works on the three-body problem

1753 Croatian astronomer Ruder Bošković claims the Moon has no atmosphere

1754 Dutch dike engineer Albert Brahms starts scientific recording of tide levels

1754 Scottish chemist Joseph Black discovers carbon dioxide

November 1, 1755 An earthquake devastates Lisbon, and prompts John Michell to develop theory of seismic waves